

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience implementing and validating predictive models across healthcare, environmental, and cryptography domains using Python, Scikit-Learn, Keras, and TensorFlow. Combines a strong mathematics and computer science foundation, including coursework in numerical optimization and NLP, with practical data engineering experience at Deutsche Bundesbank, and daily use of AI-assisted development tools (Claude Code) to accelerate model and pipeline development.

Core Competencies

- **Machine Learning & Predictive Modeling:** Random Forest, SVM, K-Means, classification, clustering, time-series analysis
- **Programming & Data:** Python (Pandas, NumPy, Scikit-Learn, Keras, TensorFlow), SQL, R, Java, C++
- **Data Engineering:** ETL/ELT pipelines, data quality automation, migration testing
- **AI-Assisted Development:** Daily use of AI coding tools (Claude Code) for pipeline and model development
- **Cloud & Tools:** Azure Cognitive Services, AWS, Tableau, Matplotlib, Seaborn, Git

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to validate business logic after data migration
 - Built automated error detection identifying statistical deviations (duplicate IDs, value-range violations) in large-scale reported banking data

Projects

- **Tecnológico de Monterrey** *10/2024–12/2024*
Predictive Modeling – Healthcare Optimization
 - Analyzed 25+ years of national health data to identify patient risk profiles and resource bottlenecks
 - Built and validated predictive models (Random Forest, Neural Networks) to forecast critical outcomes; applied K-Means clustering to segment high-risk groups
 - Ran exploratory time-series and regional analyses to uncover seasonal patterns
- **Tecnológico de Monterrey** *04/2023–06/2023*
Air Quality Analysis Mexico City – SVM Classification
 - Built SVM models (linear and RBF kernels) to classify critical ozone levels from citywide NO_x/O₃ data with high accuracy
- **Tecnológico de Monterrey** *10/2024–02/2025*
Lattice-Based Cryptography Research
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science *2022–2027*

Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: machine learning & AI, data analysis & statistics, NLP, numerical optimization.

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.

References

- Available upon request.